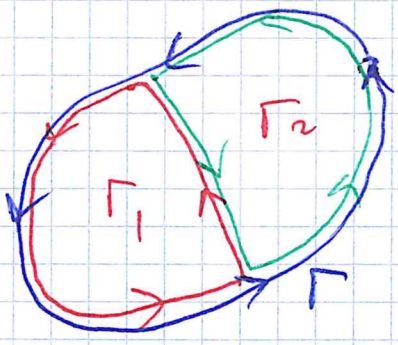
Indeed $I(\Gamma)$ takes only discrete, integer values, so changing $I(\Gamma)$ means it jumping by ± 1 or more. But deformation of Γ can be a series of tiny deformations, in which the integral for $I(\Gamma)$ would change only weakly. Which is compatible only if $I(\Gamma)$ stays constant in the process.

Note: can also see this by moving to complex plane, transforming $I(\Gamma)$ to a contour integral of an analytic function, which can have the contour deformed w/o crossing the pole.

(2) If Γ doesn't enclose FPs, $I(\Gamma)=0$ (because can deform to a tiny contour on which $\vec{f}=(f_1, f_2)$ is uniform and change of ψ is zero).

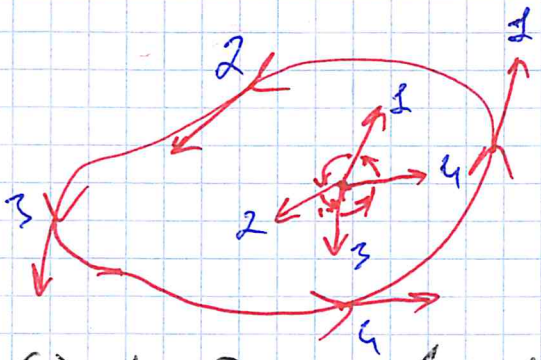


(3) $I(\Gamma)$ is additive, $I(\Gamma)=I(\Gamma_1)+I(\Gamma_2)$ if $\Gamma=\Gamma_1+\Gamma_2$ (joints cancel in the integral)



(4) Reversing time doesn't change $I(\Gamma)$, as ψ doesn't change when $f_{1,2} \to -f_{1,2}$.

(5) For a closed trajectory, e.g. a periodic orbit, $I(\Gamma)=+1$. This is since the field $\vec{f}$ is tangent to the orbit at every point.

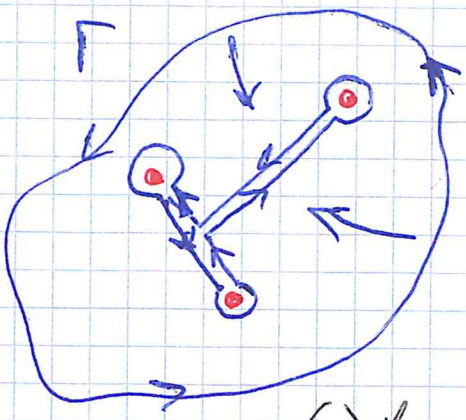


(6) The Poincaré index of an isolated FP is the Poincaré index of any simple curve encircling this FP.

(7) For a curve encircling multiple FPs index

is the sum of indices of individual FPs.

Shown by curve deformation.



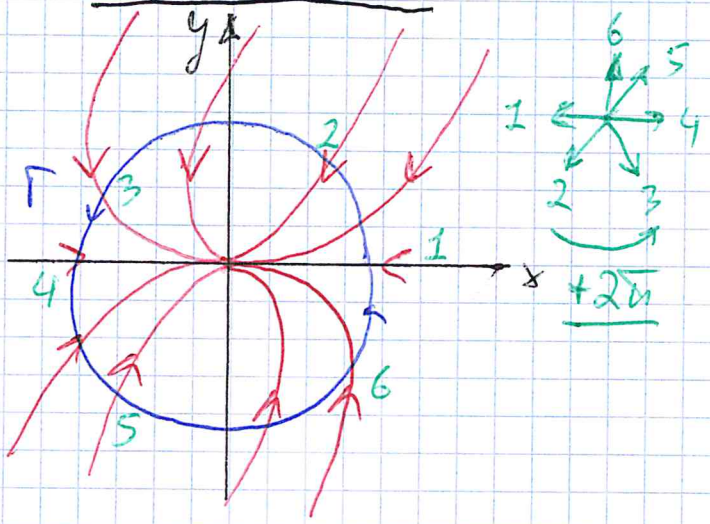
(8) Poincaré index

(*) of a hyperbolic sink/source is +1

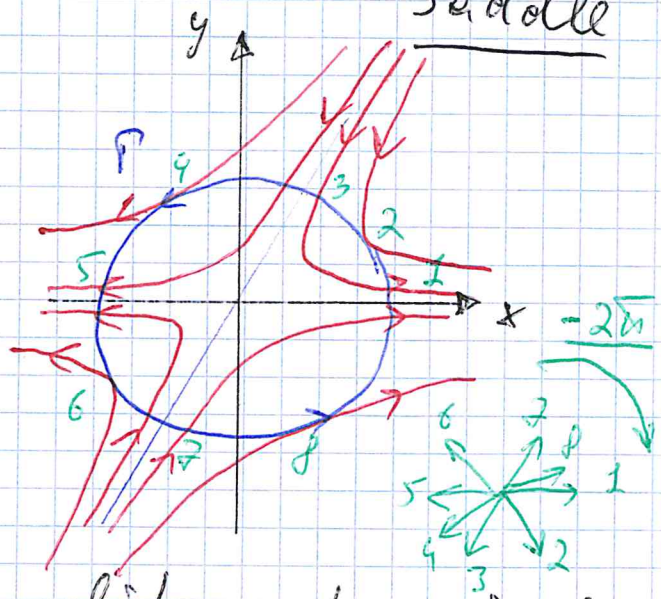
(*) of a hyperbolic saddle is −1

(*) for non-hyperbolic FPs proceed on a case by case basis (e.g. $I(\bar{x}_0)=0$ for a saddle-node).

Stable node



Saddle



Corollary: any periodic orbit must encircle one or more fixed points and the sum of their indices must be +1.

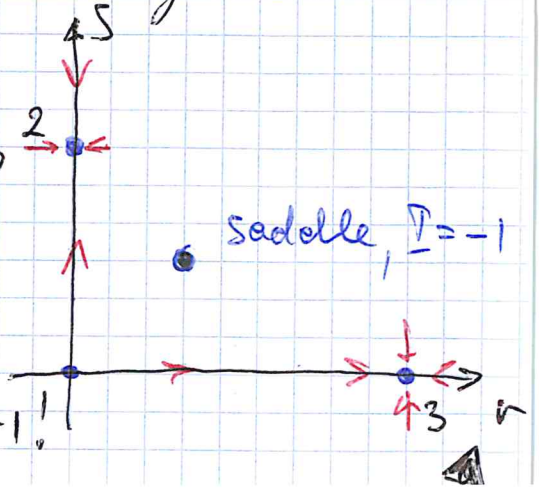
This can be used to exclude periodic orbits!

Example Consider Lotka-Volterra system

$$\dot{r} = r(3-r-2s)$$
$$\dot{s} = s(2-r-s)$$

Does it have a periodic orbit for $r,s>0$?

nodes	(0,0)	No! Since can't cross axes, can't encircle the nodes, and saddle $I=-1$!
	(3,0)	
	(0,2)	
saddle	(1,1)	



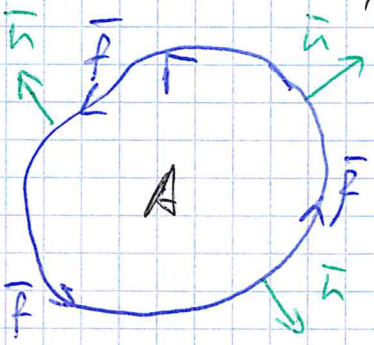
§4.2 Dulac's Criterion

Some other useful negative tests.

Theorem Dulac's theorem

If there is a continuously differentiable function $\phi(x,y)$ such that $\nabla\cdot(\phi\vec{f})>0$ (or <0) everywhere on a simply connected domain $D\subset R^2$ then there are no periodic orbits lying entirely in D.

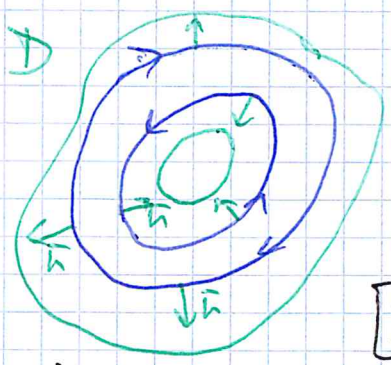
Proof Assume $\nabla\cdot(\phi\vec{f})>0$. Suppose a periodic orbit exists, let $\vec{n}$ be an outward normal to Γ



Then $0=\oint_\Gamma \phi\vec{f}\cdot\vec{n}\,dl = \iint_A \nabla\cdot(\phi\vec{f})\,dS>0$, → contradiction!

(*) In practice often use $\phi=1$, then no periodic orbit if $\nabla\cdot\vec{f}>0$ or $\nabla\cdot\vec{f}<0$ in D. This is the divergence test.

(*) If $\nabla\cdot(\vec{f}\phi)$ maintains sign in a doubly-connected domain $D\subset R^2$, then there is at most one periodic orbit in D and it must enclose the hole.



Proof: If not, consider divergence theorem between the two orbits in the same fashion, get contradiction.

Example Lotka-Volterra

$\dot r = r(a-br-cs)$

$\dot s = s(d-er-fs)$

with $b,f>0$

Can it have periodic orbits for $r,s>0$?

Take $\phi=\frac{1}{rs}$, ⇒

$$\nabla\cdot(\phi\bar{f}) = \frac{\partial}{\partial r}\left(\frac{a-br-cs}{s}\right) + \frac{\partial}{\partial s}\left(\frac{d-er-fs}{r}\right) = -\frac{b}{s} - \frac{f}{r} < 0,$$

$\Rightarrow$ no periodic orbits. Earlier example had $b=f=1$.

Can extend these concepts to R^n regarding the invariant sets of dimension $(n-1)$ using n-dimensional divergence theorem.

Example Lorenz system

$$\begin{pmatrix}\dot{x}\\ \dot{y}\\ \dot{z}\end{pmatrix} = \begin{pmatrix}\sigma(y-x)\\ rx-y-zx\\ -bz+xy\end{pmatrix}, \quad r,\sigma,b>0.$$

Here $\nabla\cdot\bar{f} = -\sigma - 1 - b < 0$, $\Rightarrow$ no invariant sets of dimension 2 enclosing non-zero volume!

But can't exclude 1D periodic orbits! ◀

Another negative test:

Theorem Gradient criterion

If there is a positive function $\rho(x,y)$: $\rho\bar{f} = \nabla\psi$ for some single-valued ψ on a simply-connected domain D, then there are no periodic orbits in D.

Proof Suppose there is a periodic orbit Γ, then

$$0 < \oint_\Gamma \rho\bar{f}\cdot d\bar{\ell} = \oint_\Gamma \nabla\psi\cdot d\bar{\ell} = 0$$

as an integral of a potential field over a closed contour. Thus, a contradiction, so no periodic orbits. ▀

Example Does the system have periodic orbits?

$$\begin{cases}\dot{x} = 2x + x^2y - y^3,\\ \dot{y} = -2y - xy^2 + x^3.\end{cases}$$

Notice that $\underbrace{e^{xy}}_{\rho}\bar{f} = \nabla\underbrace{\left[e^{xy}(x-y^2)\right]}_{\psi}$.

As $\rho>0$ and single valued, gradient criterion implies no periodic orbits. ◀

§4.3 Poincaré-Bendixson theorem

Now we describe an important result on the existence of periodic orbits.

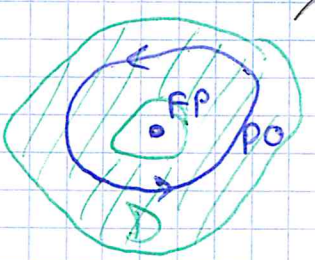
Theorem Poincaré-Bendixson

If the forward orbit $\mathcal{O}^+(\bar{x})$ of some point $\bar{x}$ remains in a compact set $D \subset \mathbb{R}^2$ that contains no fixed points, then $\omega(\bar{x})$ contains a periodic orbit.

Proof starred (*), see attached sheet.

Note: (*) true only in $\mathbb{R}^2$, no higher-D analogue (*) set D cannot be simply-connected as then a periodic orbit must encircle a FP in D by Poincaré index test (a hole in D can contain FP).

Example
$$\dot{x} = x - y - x(x^2+y^2),$$
$$\dot{y} = y + x - y(x^2+y^2)$$

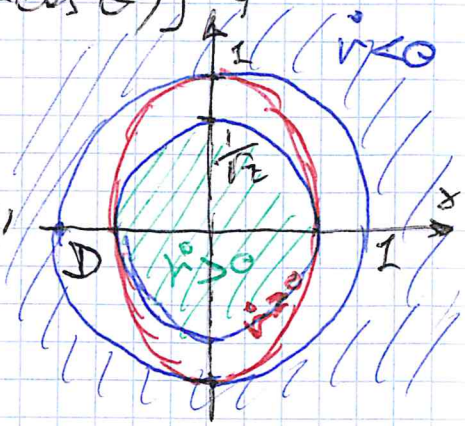


Linearized $\hat{A} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$, $\det = \tau = 2$, $\tau > 0$, $\tau^2 < 4\det$, $\Rightarrow$ unstable focus at $\bar{0}$.

But for cubic terms "push" inwards, suspect PO!

In polars $\dot{r} = \frac{x\dot{x}+y\dot{y}}{r} = r\left[1 - r^2(1+\cos^2\theta)\right]$

$\dot{r} = 0$ at $r = \frac{1}{\sqrt{1+\cos^2\theta}}$ or $\left(\frac{x}{1/\sqrt{2}}\right)^2 + y^2 = 1$.



See that $\dot{r} < 0$ for $r > 1$, $\dot{r} > 0$ for $r < \frac{1}{\sqrt{2}}$.

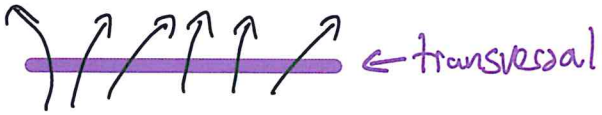
Thus for any $\bar{x}$ $\mathcal{O}^+(\bar{x})$ ends up in $D = \{\bar{x} : \frac{1}{\sqrt{2}} < r < 1\}$. Since $\dot{\theta} = 1 + \frac{1}{2}r^2\sin 2\theta > 0$ in D, have no FP in D, $\Rightarrow$ a periodic orbit must exist in D by Poincaré-Bendixson!

Bonus: Since $\nabla \cdot f = 2 - 2x^2 - 5r^2 < 0$ in D (as $< -\frac{1}{2} - 2x^2$), have only one periodic orbit by divergence test. ◭

4.3 **Sketch Proof of the Poincaré–Bendixson Theorem**

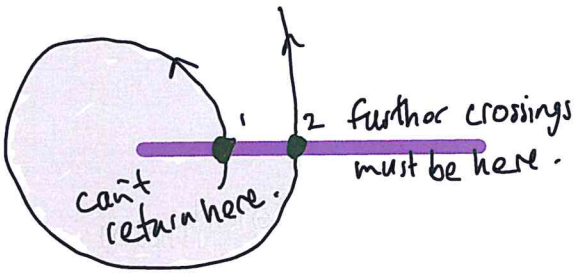
The Poincaré–Bendixson Theorem: If the forward orbit $\mathcal{O}^+(\mathbf{x})$ of some point $\mathbf{x}$ remains in a compact set $K \in \mathbb{R}^2$ that contains no fixed points then $\omega(\mathbf{x})$ is a periodic orbit.

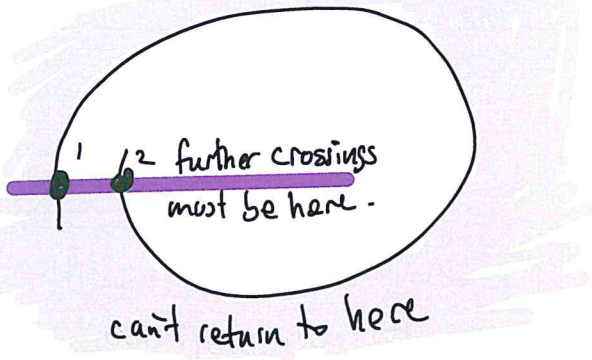
The proof is somewhat intricate, but the essentials are outlined in this handout. The key new idea within it is that of a *transversal*: a curve segment that trajectories all cross in the same direction. A transversal can be set up through any non-fixed point (e.g. choose a short straight line segment perpendicular to the flow).



Lemma: If a trajectory has multiple intersections with the transversal, then the intersections are arranged monotonically along the transversal.

See sketches below - there are only a small number of possible arrangements after two intersections, and then further intersections can only occur in one direction. Notice that this is where the $\mathbb{R}^2$ is required (Jordan curve theorem: there is an 'inside' and an 'outside' of a closed curve.)



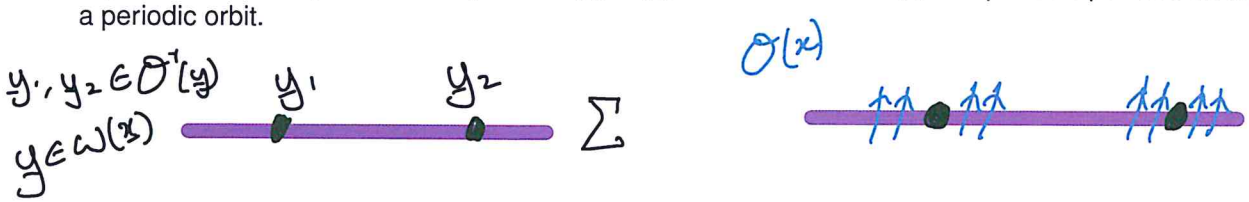


Poincaré–Bendixson sketch proof:

(1) Choose any $\mathbf{y} \in \omega(\mathbf{x})$, and then choose any $\mathbf{z} \in \omega(\mathbf{y})$. We can do this because these ω-limit sets are non-empty as K is compact. Also note that everything is in K, where there are no fixed points. We want to show that $\mathbf{y}$ is on a periodic orbit.

(2) Choose a transversal through $\mathbf{z}$ and call it Σ. As $\mathbf{z} \in \omega(\mathbf{y})$, we can find points in $\mathcal{O}^+(\mathbf{y})$ as close as we like to $\mathbf{z}$. By choosing two that are sufficiently close to $\mathbf{z}$ and running them forward or back in time a small amount, we can find $\mathbf{y}_1$ and $\mathbf{y}_2$ which are both in $\mathcal{O}^+(\mathbf{y})$ and both on Σ. (Now $\mathbf{z}$'s job is done – it was just needed to establish suitable Σ, $\mathbf{y}_1$ and $\mathbf{y}_2$.)

(3) Both $\mathbf{y}_1$ & $\mathbf{y}_2$ are in $\mathcal{O}^+(\mathbf{y})$ so are also in $\omega(\mathbf{x})$ (ω-limit sets are invariant). By a similar argument to above, we can find an infinite sequence of points in $\mathcal{O}^+(\mathbf{x})$ that are on Σ and tend to $\mathbf{y}_1$, and same for $\mathbf{y}_2$. Now use the monotonicity argument, thinking about successive crossings of $\mathcal{O}^+(\mathbf{x})$ and Σ. Intersections of $\mathcal{O}^+(\mathbf{x})$ and Σ are arranged in monotonic order (by Lemma) so cannot have sub-sequences tending to different limits. So is impossible to satisfy monotonicity unless $\mathbf{y}_1 = \mathbf{y}_2$. And we're there: $\mathbf{y}_1$ is a periodic point so actually $\mathcal{O}(\mathbf{y})$ is a periodic orbit.



[Bonus bit for proof enthusiasts: we should also check there are no other period orbits in $\omega(\mathbf{x})$, as all we can deduce so far is that there is at least one periodic orbit. This can be done by using a similar argument: no more than one point of $\omega(\mathbf{x})$ on any transversal. Can also show that ω-limit sets in compact sets are connected. Combine these to give that $\omega(\mathbf{x})$ is exactly the one periodic orbit. Details left to the reader!]